
NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY



COMPUTER COMMUNICATION AND NETWORKS (CT-376)

REPORT

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Complex Computing Problem Assessment Rubrics

Course Code: CT-376		Course Title: Computer Communication Networks	
Criteria and Scales			
Excellent (3)	Good (2)	Average (1)	Poor (0)
Criterion 1: Understanding the Problem: How well the problem statement is understood by the student			
Understands the problem clearly and identify the underlying issues and functionalities.	Adequately understands the problem and identifies the underlying issues and functionalities.	Inadequately defines the problem and identifies the underlying issues and functionalities.	Fails to define the problem adequately and does not identify the underlying issues and functionalities.
Criterion 2: Research: The amount of research that is used in solving the problem			
Contains all the information needed for solving the problem	Good research leads to a successful solution	Mediocre research which may or may not lead to an adequate solution	No apparent research
Criterion 3: Code: How complete the code is along with the assumptions?			
Complete the code according to the selected functionalities of the given case with clear assumptions	Incomplete code according to the selected functionalities of the given case with clear assumptions	Incomplete code according to the selected functionalities of the given case with unclear assumptions	Wrong code and naming conventions
Criterion 4: Report: How thorough and well organized is the solution?			
All the necessary information is organized for easy use insolving the problem	Good information organized well could lead to a good solution	Mediocre information which may or may not lead to a solution	No report provided
Criterion 5: Labeling: How well defined and labeled is the solution?			
All the necessary information is labelled (i.e. port no.) for better understanding	Good information about the topology is labelled	Incomplete label according to the selected functionalities	Not Labelled

INTRODUCTION

In this project, we have designed and implemented a sophisticated network topology for a smart city infrastructure. The objective was to integrate multiple network segments, each catering to different city services, using advanced networking concepts and technologies. This network is designed to support the seamless operation of various departments, including healthcare, security and surveillance, utilities, parking, retail, smart buildings, and residential areas.

PROBLEM STATEMENT

Design a complex network topology in which the terminologies (i.e. sub-netting, Telnet, Security (ACL), Eth-Trunk (LACP based), Static Routes, RIP, STP convergence, VLANs (ROAS), DHCP, and FTP) must be used.

PROBLEM DESCRIPTION

The objective of this project is to design and implement a comprehensive and robust network topology for a smart city infrastructure, integrating multiple departments with distinct requirements including healthcare, security and surveillance, utilities, parking, retail, smart buildings, and residential areas. The network must ensure high performance, reliability, security, and efficient management across this complex and diverse environment. Key challenges include enabling secure inter-departmental connectivity, ensuring scalability and reliability, protecting sensitive data, optimizing performance, and simplifying network administration. The solution involves using sub-netting and VLANs for segmentation, secure remote management with Telnet and ACLs, bandwidth optimization through Eth-Trunk (LACP), efficient routing with static routes and RIP, loop prevention with STP, inter-VLAN communication using Router on a Stick (ROAS), automated IP address management via DHCP, and secure file transfer with FTP. The goal is to create an integrated network that supports seamless operation of all smart city services, enhancing the efficiency, security, and resilience of the city's infrastructure.

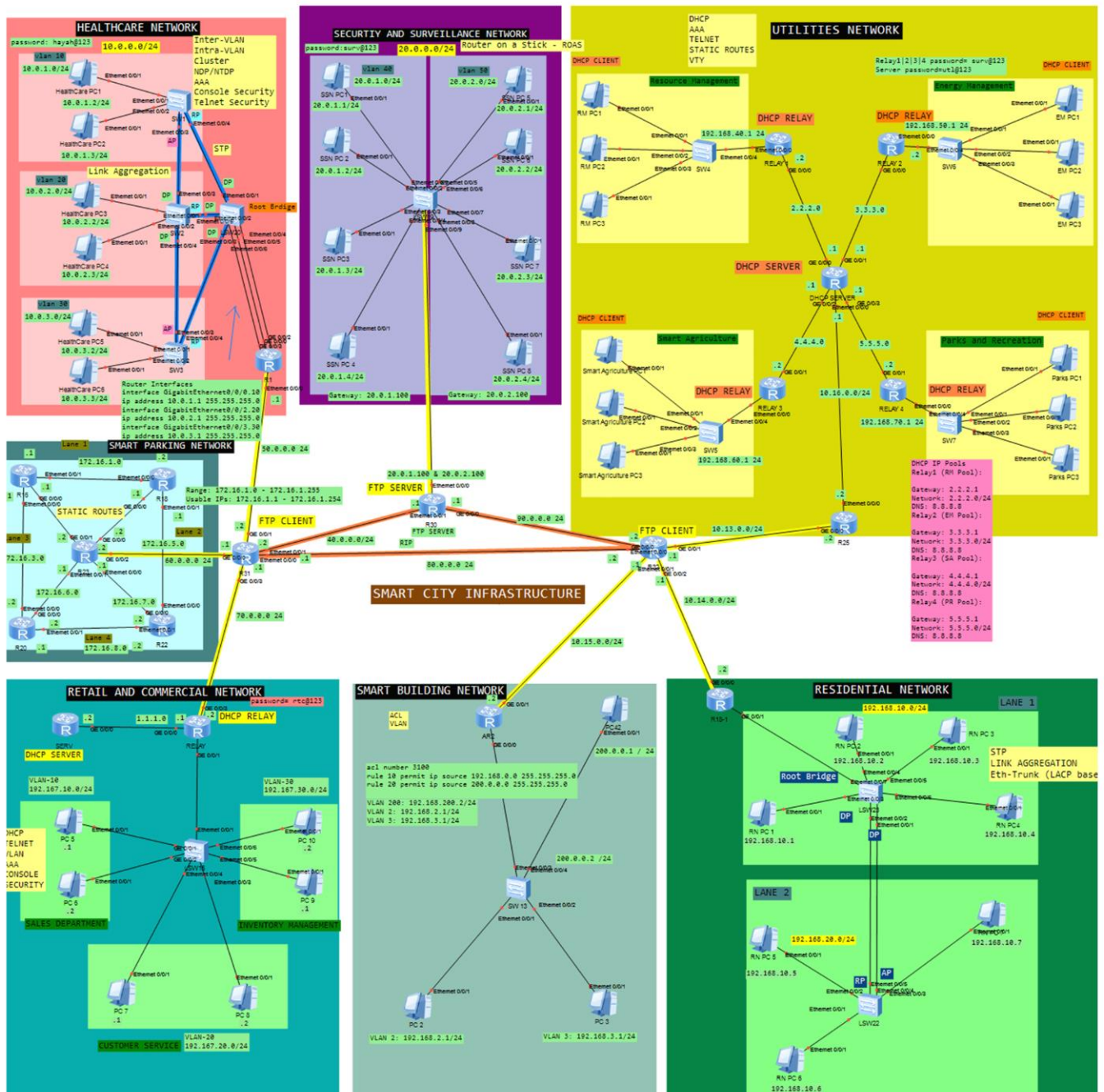
Technologies Used in the Smart City Infrastructure

1. Sub-netting
2. Telnet
3. Security (Access Control Lists - ACL)
4. Eth-Trunk (Link Aggregation Control Protocol - LACP based)
5. Static Routes
6. Routing Information Protocol (RIP)
7. Spanning Tree Protocol (STP)
8. Virtual LANs (VLANs)
9. Router on a Stick (ROAS)
10. Dynamic Host Configuration Protocol (DHCP)
11. File Transfer Protocol (FTP)
12. Open Shortest Path First (OSPF)
13. AAA (Authentication, Authorization, and Accounting)
14. Console Security
15. NDP/NDTP

PROJECT SCOPE

The topology that is being shown highlights the network architecture for a smart city infrastructure and shows how various divisions and services are connected to one another. This topic is crucial as it shows how different technologies may be combined in a harmonious way to create an infrastructure that can support the needs of a smart city. Urban planners and technologists can design more robust, effective, and flexible urban environments by comprehending and putting this topology into practice.

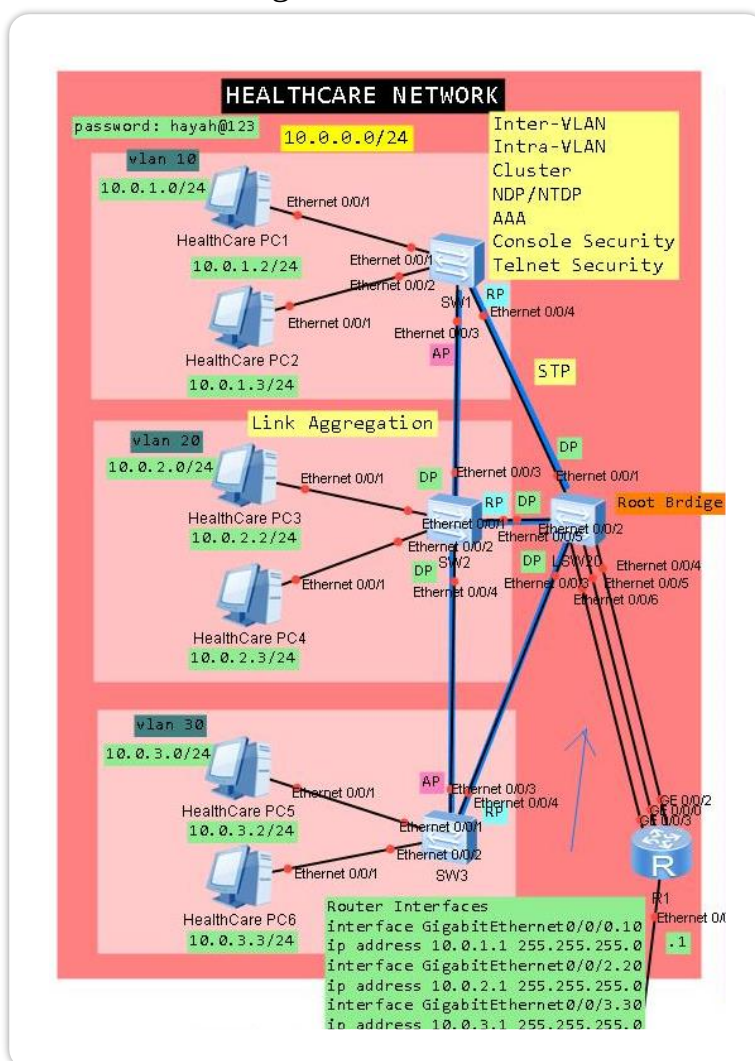
SYSTEM TOPOLOGY



SUB TOPOLOGIES

Healthcare Network

The healthcare network is designed to support the critical infrastructure needed for medical facilities within the smart city. This network is meticulously structured to ensure high security, reliability, and efficient management of healthcare services.



Purpose:

To provide a secure, reliable, and efficient network for healthcare services, including patient records, administrative data, and medical devices.

Components:

1. Routers: 1 (R1)
2. Switches: 4
LSW20 is the root bridge, the ports associated with it are Designated ports
3. Computers/PCs: 6

Technologies Implemented:

1. Inter-VLAN Routing
2. Telnet Security
3. Link Aggregation
4. Authentication, Authorization, and Accounting (AAA)
5. Console Security
6. Spanning Tree Protocol (STP)

Description:

1. **Inter-VLAN Routing:** VLANs segment the network into logical groups with routers enabling communication between them. We have developed a network which consist of 3 VLANs i.e, 10,20 and 30
2. **Telnet Security:** Remote management secured with ACLs and encrypted Telnet sessions.
3. **Link Aggregation:** Multiple switch ports combined using manual mode for increased bandwidth and redundancy.
4. **AAA:** Controls access, authorizes actions, and maintains an audit trail using services like RADIUS or TACACS+.
5. **Console Security:** Secured console access with strong passwords and idle session timeouts.
6. **STP:** Prevents network loops by ensuring a loop-free topology through configured switches. STP is applied between four switches (SW1,SW2,SW3 and LSW20) where LSW20 in the root bridge and all the ports associated with it are Designated ports and the port on SS# Ethernet 0/0/3 is the blocked port.
7. **NTDP and NDP:** These features are enabled for network topology discovery and maintenance.
- 8.

Telnet and Console Password:

-Username: admin

-Password: hayah@123

CONFIGURATION

Network (10.0.0.0/24)

- **Interface vlanif 10:** ip address 10.0.1.0/24
 - **PC1:** 10.0.1.2/24
 - **PC2:** 10.0.1.3/24
- **Interface vlanif 20:** ip address 10.0.2.0/24
 - **PC1:** 10.0.2.2/24
 - **PC2:** 10.0.2.3/24
- **Interface vlanif 30:** ip address 10.0.3.0/24
 - **PC1:** 10.0.3.2/24
 - **PC2:** 10.0.3.3/24

Router Interfaces

- **Interface GigabitEthernet0/0/0.10**
 - ip address 10.0.1.1 255.255.255.0
- **Interface GigabitEthernet0/0/2.20**
 - ip address 10.0.2.1 255.255.255.0
- **Interface GigabitEthernet0/0/3.30**
 - ip address 10.0.3.1 255.255.255.0
- **Interface GigabitEthernet0/0/1**
 - ip address 15.15.15.1 255.255.255.0

Configuration for MAIN SWITCH (LSW20)

- **System name**
 - sysname MainSwitch
- **VLAN Configuration #VLAN**
 - vlan batch 10 20 30
- **NTDP and NDP #Cluster, NTDP, NDP**
 - cluster enable
 - ntdp enable
 - ndp enable
 - drop illegal-mac alarm
 - diffserv domain default
 - drop-profile default
- **AAA #AAA**
 - AAA
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - domain default_admin
 - local-user admin password cipher hayah@123
 - local-user admin service-type http
- **Telnet Security #Telnet**
 - telnet server enable
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **Console Security #Console**
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher hayah@123
- **Interface Vlanif10 #VLAN**
 - ip address 10.0.1.1 255.255.255.0

- **Interface Vlanif20 #VLAN**
 - ip address 10.0.2.1 255.255.255.0
- **Interface Vlanif30 #VLAN**
 - ip address 10.0.3.1 255.255.255.0
- **STP #STP**
 - stp enable
 - stp mode rstp
 - stp priority 0
 - stp root primary
- save

Configuration for ACCESS SWITCH1 (SW1) - VLAN 10

- **System name**
 - sysname AccessSwitch1
- **VLAN Configuration #VLAN**
 - vlan batch 10
- **Cluster, NTDP, NDP #Cluster, NTDP, NDP**
 - cluster enable
 - ntdp enable
 - ndp enable
 - drop illegal-mac alarm
 - diffserv domain default
 - drop-profile default
- **AAA #AAA**
 - AAA
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - domain default_admin
 - local-user admin password cipher hayah@123
 - local-user admin service-type http
- **Telnet Security #Telnet**
 - telnet server enable
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **Console Security #Console**
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher hayah@123
- **Interface Vlanif10 #VLAN**
 - ip address 10.0.1.1 255.255.255.0
- **STP #STP**
 - stp enable
 - stp mode rstp
 - stp root secondary
- save

Configuration for ACCESS SWITCH2 (SW2) - VLAN 20

- **System name**
 - sysname AccessSwitch2
- **VLAN Configuration #VLAN**
 - vlan batch 20
- **Cluster, NTDP, NDP #Cluster, NTDP, NDP**
 - cluster enable
 - ntdp enable
 - ndp enable
 - drop illegal-mac alarm

- o diffserv domain default
 - o drop-profile default
- **AAA #AAA**
 - o AAA
 - o authentication-scheme default
 - o authorization-scheme default
 - o accounting-scheme default
 - o domain default
 - o domain default_admin
 - o local-user admin password cipher hayah@123
 - o local-user admin service-type http
- **Telnet Security #Telnet**
 - o telnet server enable
 - o **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **Console Security #Console**
 - o **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher hayah@123
- **Interface Vlanif20 #VLAN**
 - o ip address 10.0.2.1 255.255.255.0
- **STP #STP**
 - o stp enable
 - o stp mode rstp
 - o stp priority 32768

Configuration for ACCESS SWITCH3 (SW3) - VLAN 30

- **System name**
 - o sysname AccessSwitch3
- **VLAN Configuration #VLAN**
 - o vlan batch 30
- **Cluster, NTDP, NDP #Cluster, NTDP, NDP**
 - o cluster enable
 - o ntdp enable
 - o ndp enable
 - o drop illegal-mac alarm
 - o diffserv domain default
 - o drop-profile default
- **AAA #AAA**
 - o AAA
 - o authentication-scheme default
 - o authorization-scheme default
 - o accounting-scheme default
 - o domain default
 - o domain default_admin
 - o local-user admin password cipher hayah@123
 - o local-user admin service-type http
- **Telnet Security #Telnet**
 - o telnet server enable
 - o **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **Console Security #Console**
 - o **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher hayah@123
- **Interface Vlanif30 #VLAN**
 - o ip address 10.0.3.1 255.255.255.0
- **STP #STP**

- o stp enable
- o stp mode rstp
- o stp priority 32768

Configuration for Router (R1)

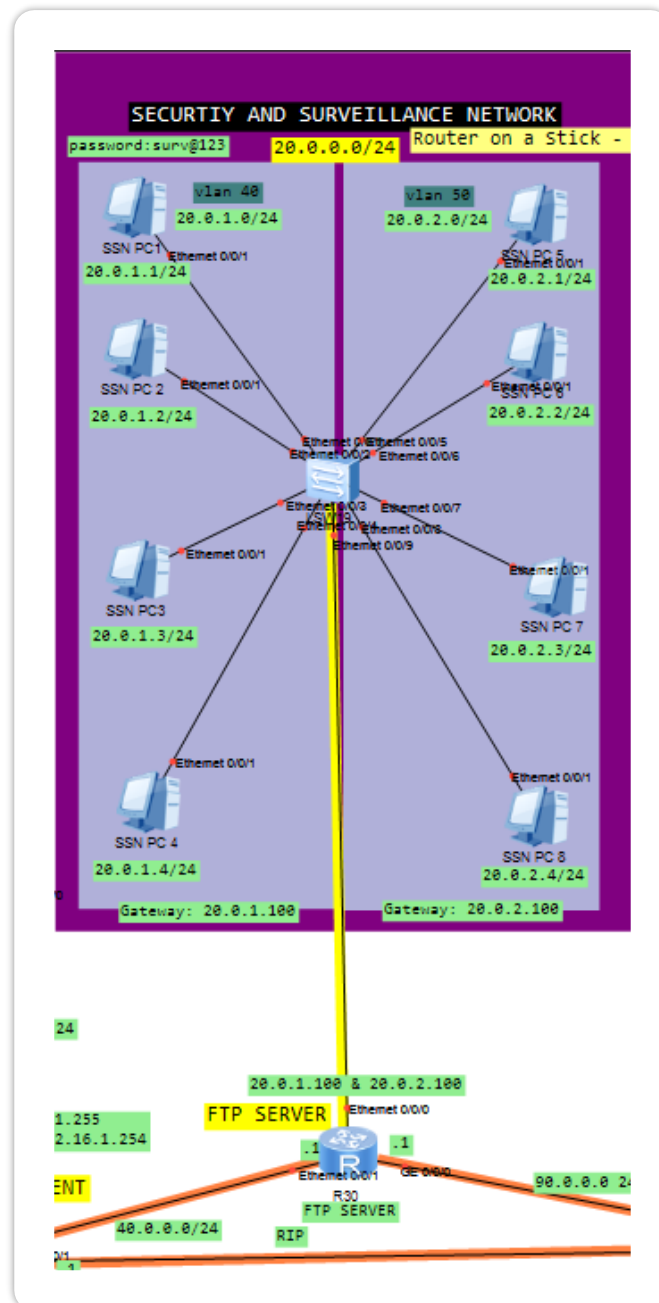
- **System name**
 - o sysname R1
- **Inter-VLAN Routing #Inter-VLAN Routing**
 - o **Interface GigabitEthernet0/0/0.10**
 - dot1q termination vid 10
 - ip address 10.0.1.1 255.255.255.0
 - o **Interface GigabitEthernet0/0/2.20**
 - dot1q termination vid 20
 - ip address 10.0.2.1 255.255.255.0
 - o **Interface GigabitEthernet0/0/3.30**
 - dot1q termination vid 30
 - ip address 10.0.3.1 255.255.255.0
 - o **Interface GigabitEthernet0/0/3.30**
 - dot1q termination vid 30
 - ip address 10.0.3.1 255.255.255.0
 - o **Interface GigabitEthernet0/0/0**
 - undo shutdown
 - o **Interface GigabitEthernet0/0/2**
 - undo shutdown
 - o **Interface GigabitEthernet0/0/3**
 - undo shutdown
 - o **Interface GigabitEthernet0/0/1**
 - ip address 15.15.15.1 255.255.255.0

JUSTIFICATION:

In healthcare networks, essential technologies ensure security, performance, and reliability. Inter-VLAN routing segments traffic for enhanced security and manageability. Telnet security restricts remote access to protect sensitive data. Link aggregation boosts network performance and reliability. AAA provides controlled access through a comprehensive security framework. Console security prevents unauthorized access to network devices. STP maintains a stable and resilient network by preventing loops. Together, these technologies support the smooth and secure operation of healthcare facilities.

SECURTIY AND SURVEILLANCE NETWORK

The security and surveillance network is meticulously designed to provide comprehensive protection and monitoring for both the digital and physical aspects of the organization. This network integrates advanced security measures, monitoring tools, and management protocols to ensure the highest levels of security, reliability, and efficiency in safeguarding the organization's resources.



Purpose:

The security and surveillance system is designed to protect the network from unauthorized access, monitor network traffic for anomalies, ensure data integrity, and provide real-time alerts for any suspicious activities. It also ensures that only authorized personnel have access to network resources and that all actions are logged for audit purposes.

Components:

- Routers: 1 (R20)
- Switches: 1
- Computers/PCs: 8

Technologies Implemented:

1. Router On a stick -(ROAS)
2. Authentication, Authorization, and Accounting (AAA)
3. Telnet Security
4. Console Security

Description:

Router On a stick -(ROAS): The Router's interface is divided into sub-interfaces, which acts as a default gateway to their respective VLANs. In this network, there is a router and a switch and some end hosts. 2 different VLANs have been created on the switch. The router's interface is divided into 2 sub-interfaces (as there are 2 different VLANs) which will acts as a default gateway to their respective VLANs. Then router will perform Inter VLAN Routing and the VLANs will communicate with each other.

Telnet and Console Password:

- Username: admin
- Password: surv@123

CONFIGURATION

Network (20.0.0.0/24)

- **Interface vlan 40:** ip address 20.0.1.0/24
 - **PC1:** 20.0.1.1/24
 - **PC2:** 20.0.1.2/24
 - **PC3:** 20.0.1.3/24
 - **PC4:** 20.0.1.4/24
- **Interface vlanif 50:** ip address 20.0.2.0/24
 - **PC1:** 20.0.2.1/24
 - **PC2:** 20.0.2.2/24
 - **PC3:** 20.0.2.3/24
 - **PC4:** 20.0.2.4/24

Router Interfaces

- **Interface Ethernet0/0/0.10**
 - ip address 20.0.1.100 255.255.255.0
- **Interface Ethernet0/0/0.20**
 - ip address 20.0.2.100 255.255.255.0

Configuration for SWITCH (LSW4)

- **System name**
 - sysname SSN_Switch
- **VLAN Configuration #VLAN**
 - vlan batch 40 50
- **Configure access ports for VLAN 40 #VLAN**
 - **Interface Ethernet0/0/1**
 - port link-type access
 - port default vlan 40
 - **Interface Ethernet0/0/2**
 - port link-type access
 - port default vlan 40
 - **Interface Ethernet0/0/3**
 - port link-type access
 - port default vlan 40
 - **Interface Ethernet0/0/4**
 - port link-type access
 - port default vlan 40
- **Configure access ports for VLAN 50 #VLAN**
 - **Interface Ethernet0/0/5**
 - port link-type access
 - port default vlan 50
 - **Interface Ethernet0/0/6**
 - port link-type access
 - port default vlan 50
 - **Interface Ethernet0/0/7**
 - port link-type access
 - port default vlan 50
 - **Interface Ethernet0/0/8**
 - port link-type access
 - port default vlan 50
- **Configure trunk port #VLAN**
 - **Interface Ethernet0/0/9**
 - port link-type trunk
 - port trunk allow-pass vlan 40 50
- **Enable telnet server #Telnet**
 - telnet server enable
- **Local user configuration #AAA**
 - local-user admin password cipher surv@123
 - local-user admin service-type telnet
- **User interface configuration for telnet #Telnet**
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **User interface configuration for console #Console**
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **AAA configuration #AAA**
 - aaa
 - authentication-scheme default
 - authorization-scheme default

- o accounting-scheme default
- o domain default_admin

Configuration for ROUTER (R6)

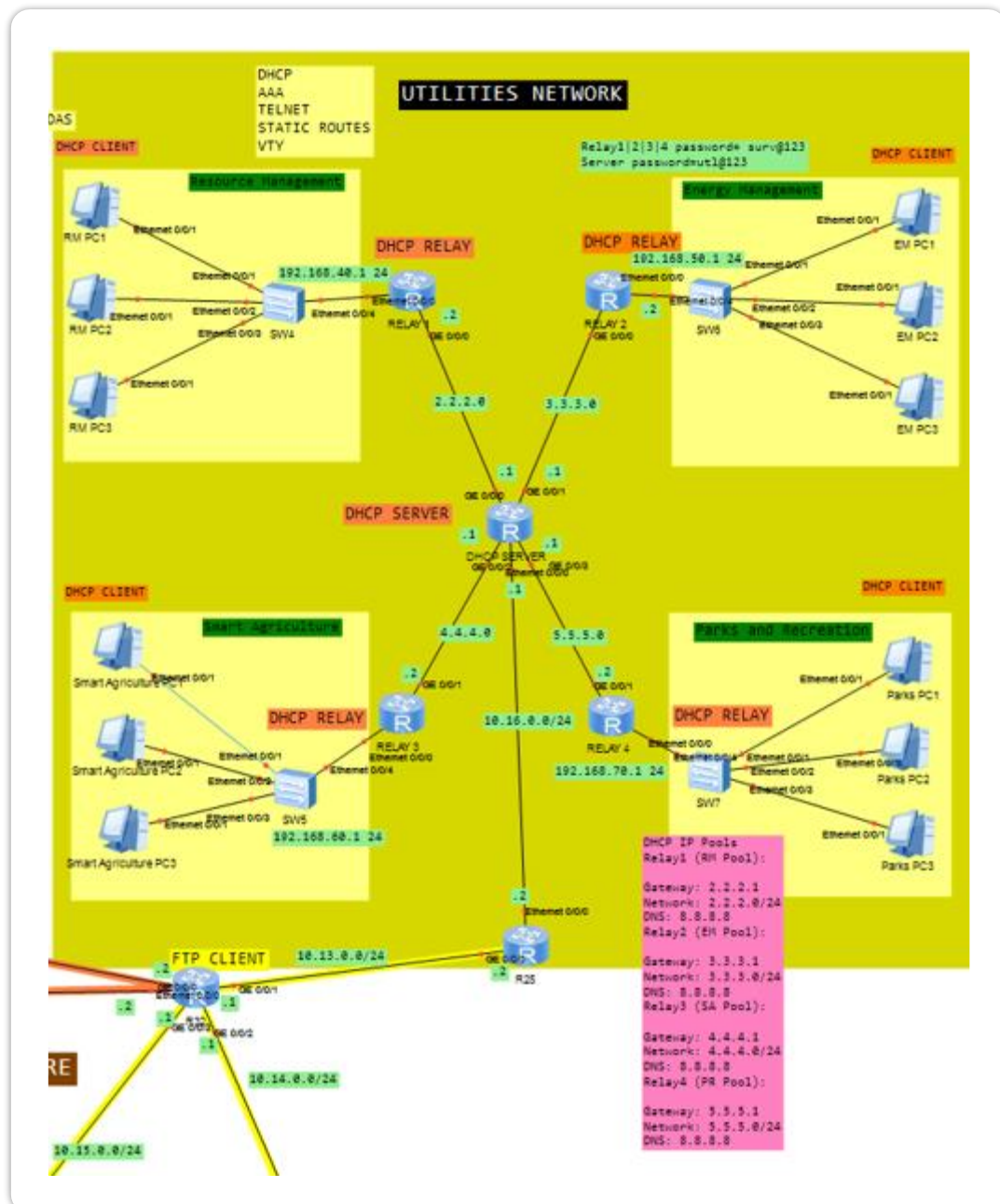
- **System name**
 - o sysname SSN_Router
- **Sub-interface for VLAN 40 #VLAN**
 - o **Interface Ethernet0/0/0.40**
 - dot1q termination vid 40
 - ip address 20.0.1.100 255.255.255.0
 - arp broadcast enable
- **Sub-interface for VLAN 50 #VLAN**
 - o **Interface Ethernet0/0/0.50**
 - dot1q termination vid 50
 - ip address 20.0.2.100 255.255.255.0
 - arp broadcast enable
- **Enable telnet server #Telnet**
 - o telnet server enable
- **Local user configuration #AAA**
 - o local-user admin password cipher surv@123
 - o local-user admin service-type telnet
- **User interface configuration for telnet #Telnet**
 - o **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
- **User interface configuration for console #Console**
 - o **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **AAA configuration #AAA**
 - o aaa
 - o authentication-scheme default
 - o authorization-scheme default
 - o accounting-scheme default
 - o domain default_admin

JUSTIFICATION:

Router on a Stick (ROAS) efficiently uses a single router interface divided into sub-interfaces to manage multiple VLANs, enabling inter-VLAN routing and simplifying network management. Telnet, however, is insecure as it transmits data in plain text; using SSH instead is recommended. Implementing AAA (Authentication, Authorization, and Accounting) enhances security by ensuring that only authorized users can access the network and their actions are monitored.

UTILITIES NETWORK

The purpose of your utilities network is to manage and monitor various network elements efficiently. It ensures smooth communication between devices, automates IP address assignment, and allows for remote management and monitoring. This setup is crucial for maintaining reliable and efficient operations within your utilities infrastructure.



Purpose:

The purpose of your utilities network is to ensure efficient management and monitoring of various network components, facilitating smooth communication between devices, automated IP address assignment, remote configuration

Components:

1. -Routers (6)
2. -Switches (4)
3. -DHCP servers (1)
4. -DHCP clients (4)
5. -DHCP Relay (4)

Technologies Implemented:

1. DHCP (Dynamic Host Configuration Protocol)
2. AAA (Authentication, Authorization, and Accounting)
3. Telnet Security
4. Static Routes
5. VTY (Virtual Teletype)

Description:

1. **DHCP (Dynamic Host Configuration Protocol):** Configure a DHCP server to manage a pool of IP addresses. Devices (DHCP clients) send a request to the DHCP server, which assigns an available IP address to each device.

DHCP IP Pools

Relay1 (RM Pool):

Gateway: 2.2.2.1

Network: 2.2.2.0/24

DNS: 8.8.8.8

Relay2 (EM Pool):

Gateway: 3.3.3.1

Network: 3.3.3.0/24

DNS: 8.8.8.8

Relay3 (SA Pool):

Gateway: 4.4.4.1

Network: 4.4.4.0/24

DNS: 8.8.8.8

Relay4 (PR Pool):

Gateway: 5.5.5.1

Network: 5.5.5.0/24

DNS: 8.8.8.8

2. **AAA (Authentication, Authorization, and Accounting):** Set up a centralized AAA server (e.g., RADIUS or TACACS+). Configure network devices to communicate with the AAA server for verifying user credentials, determining access levels, and logging user activities.
3. **Telnet Security:** Enable Telnet on network devices to allow remote access. For enhanced security, configure SSH (Secure Shell) instead of Telnet to encrypt communication.
4. **Static Routes:** Manually configure static routes on routers. Specify the destination network and the next-hop address to ensure predictable and efficient data flow.
5. **VTY (Virtual Teletype):** Configure VTY lines on network devices to accept remote connections via Telnet or SSH. This allows administrators to log in and manage the devices remotely.

CONFIGURATION

DHCP Server Configuration

- **System name**
 - `sysname dhcp server`
- **AAA Configuration #AAA**
 - `aaa`
 - `authentication-scheme default`
 - `authorization-scheme default`
 - `accounting-scheme default`
 - `domain default`
 - `domain default_admin`
 - `local-user admin password cipher utl@123`
 - `local-user admin service-type http`
 - `telnet server enable`
- **User Interface Configuration #Telnet**
 - **User-interface vty 0 4**
 - `authentication-mode aaa`
 - `protocol inbound telnet`
 - `user privilege level 15`
 - **User-interface console 0**
 - `authentication-mode password`
 - `set authentication password cipher utl@123`
- **Interface Configuration**
 - **Interface GigabitEthernet 0/0/0**
 - `ip address 2.2.2.1 255.255.255.0`
 - **Interface GigabitEthernet 0/0/1**
 - `ip address 3.3.3.1 255.255.255.0`
 - **Interface GigabitEthernet 0/0/2**
 - `ip address 4.4.4.1 255.255.255.0`
 - **Interface GigabitEthernet 0/0/3**
 - `ip address 5.5.5.1 255.255.255.0`
 - **Interface NULL0**
 - `ip route-static 0.0.0.0 0.0.0.0 2.2.2.2`
 - `dhcp enable`
- **DHCP Pools Configuration #DHCP**
 - **IP Pool RM**
 - `network 2.2.2.0 mask 255.255.255.0`
 - `gateway-list 2.2.2.1`
 - `dns-list 8.8.8.8`
 - **IP Pool EM**

- network 3.3.3.0 mask 255.255.255.0
- gateway-list 3.3.3.1
- dns-list 8.8.8.8
- **IP Pool SA**
 - network 4.4.4.0 mask 255.255.255.0
 - gateway-list 4.4.4.1
 - dns-list 8.8.8.8
- **IP Pool PR**
 - network 5.5.5.0 mask 255.255.255.0
 - gateway-list 5.5.5.1
 - dns-list 8.8.8.8
- save

Relay1 Configuration

- **System name**
 - sysname Relay1
- **AAA Configuration #AAA**
 - aaa
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - domain default_admin
 - local-user admin password cipher surv@123
 - local-user admin service-type http
 - telnet server enable
- **User Interface Configuration #Telnet**
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **Interface Configuration**
 - **Interface Ethernet0/0/0**
 - ip address 192.168.40.1 255.255.255.0
 - **Interface GigabitEthernet 0/0/0**
 - ip address 2.2.2.2 255.255.255.0
 - **Interface NULL0**
 - ip route-static 0.0.0.0 0.0.0.0 2.2.2.1
 - dhcp enable
 - **Interface Ethernet0/0/0**
 - dhcp select relay
 - dhcp relay server-ip 2.2.2.1
- save

Relay2 Configuration

- **System name**
 - sysname Relay2
- **AAA Configuration #AAA**
 - aaa
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - domain default_admin
 - local-user admin password cipher surv@123
 - local-user admin service-type http
 - telnet server enable

- **User Interface Configuration #Telnet**
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **Interface Configuration**
 - **Interface Ethernet0/0/0**
 - ip address 192.168.50.1 255.255.255.0
 - **Interface GigabitEthernet 0/0/0**
 - ip address 3.3.3.2 255.255.255.0
 - **Interface NULL0**
 - ip route-static 0.0.0.0 0.0.0.0 3.3.3.1
 - dhcp enable
 - **Interface Ethernet0/0/0**
 - dhcp select relay
 - dhcp relay server-ip 3.3.3.1
- save

Relay3 Configuration

- **System name**
 - sysname Relay3
- **AAA Configuration #AAA**
 - aaa
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - domain default_admin
 - local-user admin password cipher surv@123
 - local-user admin service-type http
 - telnet server enable
- **User Interface Configuration #Telnet**
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **Interface Configuration**
 - **Interface Ethernet0/0/0**
 - ip address 192.168.60.1 255.255.255.0
 - **Interface GigabitEthernet 0/0/0**
 - ip address 4.4.4.2 255.255.255.0
 - **Interface NULL0**
 - ip route-static 0.0.0.0 0.0.0.0 4.4.4.1
 - dhcp enable
 - **Interface Ethernet0/0/0**
 - dhcp select relay
 - dhcp relay server-ip 4.4.4.1
- save

Relay4 Configuration

- **System name**
 - sysname Relay4
- **AAA Configuration #AAA**
 - aaa

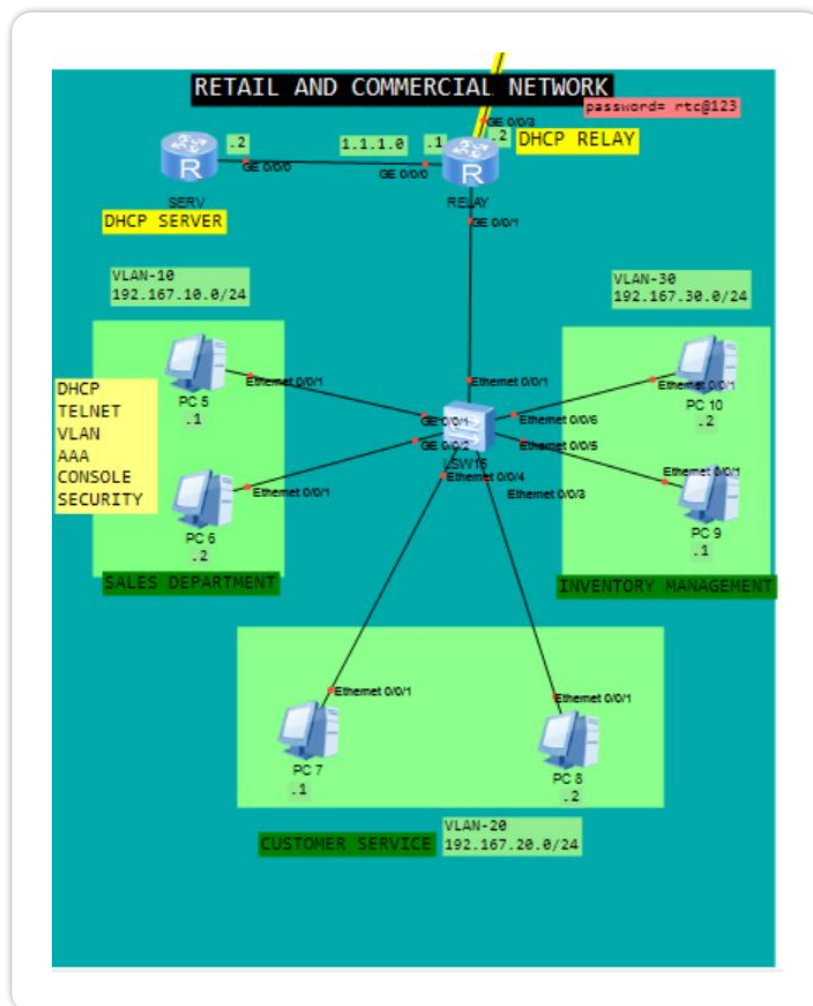
- authentication-scheme default
- authorization-scheme default
- accounting-scheme default
- domain default
- domain default_admin
- local-user admin password cipher surv@123
- local-user admin service-type http
- telnet server enable
- **User Interface Configuration #Telnet**
 - **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
 - **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher surv@123
- **Interface Configuration**
 - **Interface Ethernet0/0/0**
 - ip address 192.168.70.1 255.255.255.0
 - **Interface GigabitEthernet 0/0/0**
 - ip address 5.5.5.2 255.255.255.0
 - **Interface NULL0**
 - ip route-static 0.0.0.0 0.0.0.0 5.5.5.1
 - dhcp enable
 - **Interface Ethernet0/0/0**
 - dhcp select relay
 - dhcp relay server-ip 5.5.5.1

JUSTIFICATION:

Implementing DHCP, AAA, Telnet Security, Static Routes, and VTY in our utilities network ensures efficient and secure management of network resources. DHCP automates IP address assignment, simplifying network administration. AAA provides robust security by controlling access and tracking user activities. Telnet (or preferably SSH) allows for remote management of devices, enhancing flexibility. Static routes ensure predictable and efficient data flow, while VTY lines enable remote configuration and monitoring. Together, these technologies create a reliable, secure, and manageable network infrastructure, essential for supporting various services and applications within your utilities network.

RETAIL AND COMMERCIAL NETWORK

This network represents the structured layout of a retail and commercial network designed to support various business operations. The network is segmented into different departments such as Sales, Inventory, and Customer Service, each with its own VLAN configuration. This segmentation ensures efficient resource management, secure communication, and streamlined operations across the organization. The network includes essential components like routers, switches, DHCP servers, and PCs, all interconnected to provide robust and reliable connectivity.



Purpose:

The purpose of this network is to facilitate communication between different departments, manage resources efficiently, and provide services such as Dynamic Host Configuration Protocol (DHCP) for assigning IP addresses automatically within the network.

Components:

1. DHCP Server: 1
2. DHCP Relay: 1
3. Routers: 2
4. Switches: 1
5. PCs: 6

Technologies Implemented:

1. DHCP (Dynamic Host Configuration Protocol)
2. Telnet Security
3. VLAN (Virtual Local Area Network)
4. AAA (Authentication, Authorization, and Accounting)
5. Console Security

Description

1. **DHCP (Dynamic Host Configuration Protocol):** Configured to manage a pool of IP addresses, automatically assigning them to devices.
2. **Telnet:** Enabled on network devices for remote access. For enhanced security, SSH (Secure Shell) is often used instead to encrypt communication.
3. **VLAN (Virtual Local Area Network):** Partitioning the physical network into multiple logical networks, each department has its own VLAN for security and management purposes.
4. **AAA (Authentication, Authorization, and Accounting):** Set up a centralized server (e.g., RADIUS or TACACS+) to verify user credentials, determine access levels, and log user activities.
5. **Console:** Provides direct access to network devices for configuration and troubleshooting.

Telnet and Console Password:

-Username: admin

-Password: rtc@123

CONFIGURATION

LSW20 Configuration

VLAN Configuration

- `vlan batch 20 30 40`

Port Configuration for GigabitEthernet 0/0/1 to GigabitEthernet 0/0/2

- `port group-member GigabitEthernet 0/0/1 to GigabitEthernet 0/0/2`
- `port link-type access`
- `port default vlan 10`
- `q`

Port Configuration for Ethernet 0/0/5 to Ethernet 0/0/6

- `port group-member Ethernet 0/0/5 to Ethernet 0/0/6`
- `port link-type access`
- `port default vlan 40`
- `q`

Port Configuration for Ethernet 0/0/3 to Ethernet 0/0/4

- `port group-member Ethernet 0/0/3 to Ethernet 0/0/4`
- `port link-type access`
- `port default vlan 30`
- `q`

Interface Configuration for Ethernet 0/0/1

- `interface Ethernet 0/0/1`
- `port link-type trunk`
- `port trunk allow-pass vlan 20 30 40`
- `dhcp relay`

DHCP Server Configuration

System Name

- `sysname DHCP SERVER`

Enable DHCP

- `dhcp enable`

Interface Configuration for GigabitEthernet 0/0/0

- `interface GigabitEthernet 0/0/0`
- `ip address 1.1.1.2 255.255.255.0`
- `dhcp select global`
- `q`

DHCP IP Pool Configuration for VLAN 20

- `ip pool LAN-20`
- `network 192.168.20.0 mask 255.255.255.0`
- `gateway-list 192.168.20.1`

- dns-list 8.8.8.8
- q

DHCP IP Pool Configuration for VLAN 30

- ip pool LAN-30
- network 192.168.30.0 mask 255.255.255.0
- gateway-list 192.168.30.1
- dns-list 8.8.8.8
- q

DHCP IP Pool Configuration for VLAN 40

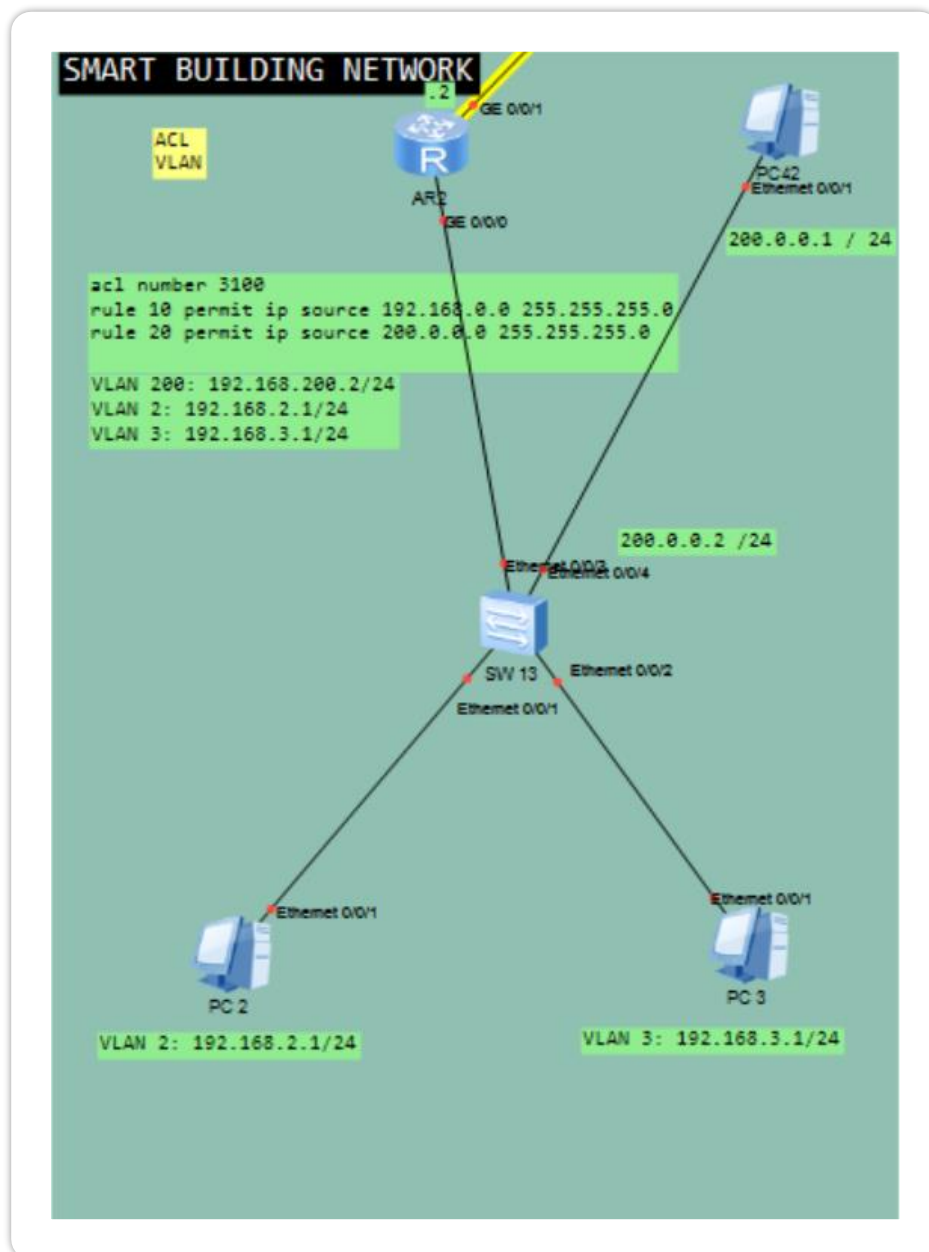
- ip pool LAN-40
- network 192.168.40.0 mask 255.255.255.0
- gateway-list 192.168.40.1
- dns-list 8.8.8.8
- q

JUSTIFICATION:

Implementing these technologies ensures that each department can operate independently within its own secure segment while still being part of the larger corporate structure. This setup enhances security by segregating traffic and allows for centralized management through protocols like Telnet/SSH and AAA services. Overall, it creates a reliable, secure, and manageable network infrastructure essential for supporting various services and applications within your retail and commercial network

SMART BUILDING NETWORK

This network represents the structured layout of a retail and commercial network within a smart building environment. The network is segmented into different departments, each with its own VLAN configuration, ensuring efficient resource management, secure communication, and streamlined operations.



Purpose:

The purpose of this network is to facilitate communication between different departments, manage resources efficiently, and provide services such as Dynamic Host Configuration Protocol (DHCP) for automatic IP address assignment

Components

1. Routers:1
2. Switches:1
3. PCs:3
4. ACL (Access Control List)
5. VLANs (Virtual Local Area Networks)

Technologies Implemented

1. ACL (Access Control List)
2. VLAN (Virtual Local Area Network)

Description:

1. **ACL (Access Control List):** Provides an additional layer of security by defining who is allowed to access the network resources based on IP addresses. In your network, ACLs are applied to routers to permit or deny traffic from specific IP ranges, enhancing security by controlling access to sensitive areas.
2. **VLAN (Virtual Local Area Network):** Enhances performance by segmenting networks into smaller parts without requiring separate physical infrastructure. Each VLAN in your network has its own IP address range, allowing specific groups or services to communicate more efficiently and securely.

CONFIGURATION

Switch Configuration

VLAN Configuration

- `vlan batch 2 3 200`

Interface Configuration

- **Interface Ethernet 0/0/1**
 - `port link-type access`
 - `port default vlan 2`
 - `quit`
- **Interface Ethernet 0/0/3**
 - `port link-type access`
 - `port default vlan 3`
 - `quit`
- **Interface Vlanif 2**
 - `ip address 192.168.2.1 255.255.255.0`
 - `quit`
- **Interface Vlanif 3**
 - `ip address 192.168.3.1 255.255.255.0`
 - `quit`

- **Interface GigabitEthernet 0/0/1**
 - port link-type access
 - port default vlan 200
 - quit
- **Interface Vlanif 200**
 - ip address 192.168.200.2 255.255.255.0
 - quit

Static Route Configuration

- ip route-static 0.0.0.0 0.0.0.0 192.168.200.1

ACL Configuration

- acl number 3100
 - rule 10 permit ip source 192.168.0.0 255.255.255.0
 - rule 20 deny ip
 - quit

Traffic Filter Configuration

- **Interface GigabitEthernet 0/0/1**
 - traffic-filter inbound acl 3100
 - quit
- **Interface Ethernet 0/0/1**
 - traffic-filter inbound acl 3100
 - quit
- **Interface Ethernet 0/0/2**
 - traffic-filter inbound acl 3100
 - quit

Telnet Server Configuration

- telnet server enable

User Interface Configuration

- **User-interface vty 0 4**
 - authentication-mode aaa
 - protocol inbound telnet
 - user privilege level 15
 - quit
- **User-interface console 0**
 - authentication-mode password
 - set authentication password cipher smart@123
 - quit

AAA Configuration

- aaa
 - authentication-scheme default
 - authorization-scheme default
 - accounting-scheme default
 - domain default
 - local-user admin password cipher smart@123
 - local-user admin service-type telnet

Router Configuration

Interface Configuration

- **Interface Ethernet 0/0/0**

- o ip address 192.168.200.1 255.255.255.0
- o quit

ACL Configuration

- acl number 3100
 - o rule 10 permit ip source 192.168.0.0 255.255.255.0
 - o rule 20 permit ip source 200.0.0.0 255.255.255.0
 - o rule 20 deny ip
 - o quit

Traffic Filter Configuration

- **Interface Ethernet 0/0/1**
 - o traffic-filter inbound acl 3100
 - o quit

Telnet Server Configuration

- telnet server enable

User Interface Configuration

- **User-interface vty 0 4**
 - o authentication-mode aaa
 - o protocol inbound telnet
 - o user privilege level 15
 - o quit
- **User-interface console 0**
 - o authentication-mode password
 - o set authentication password cipher smart@123
 - o quit

AAA Configuration

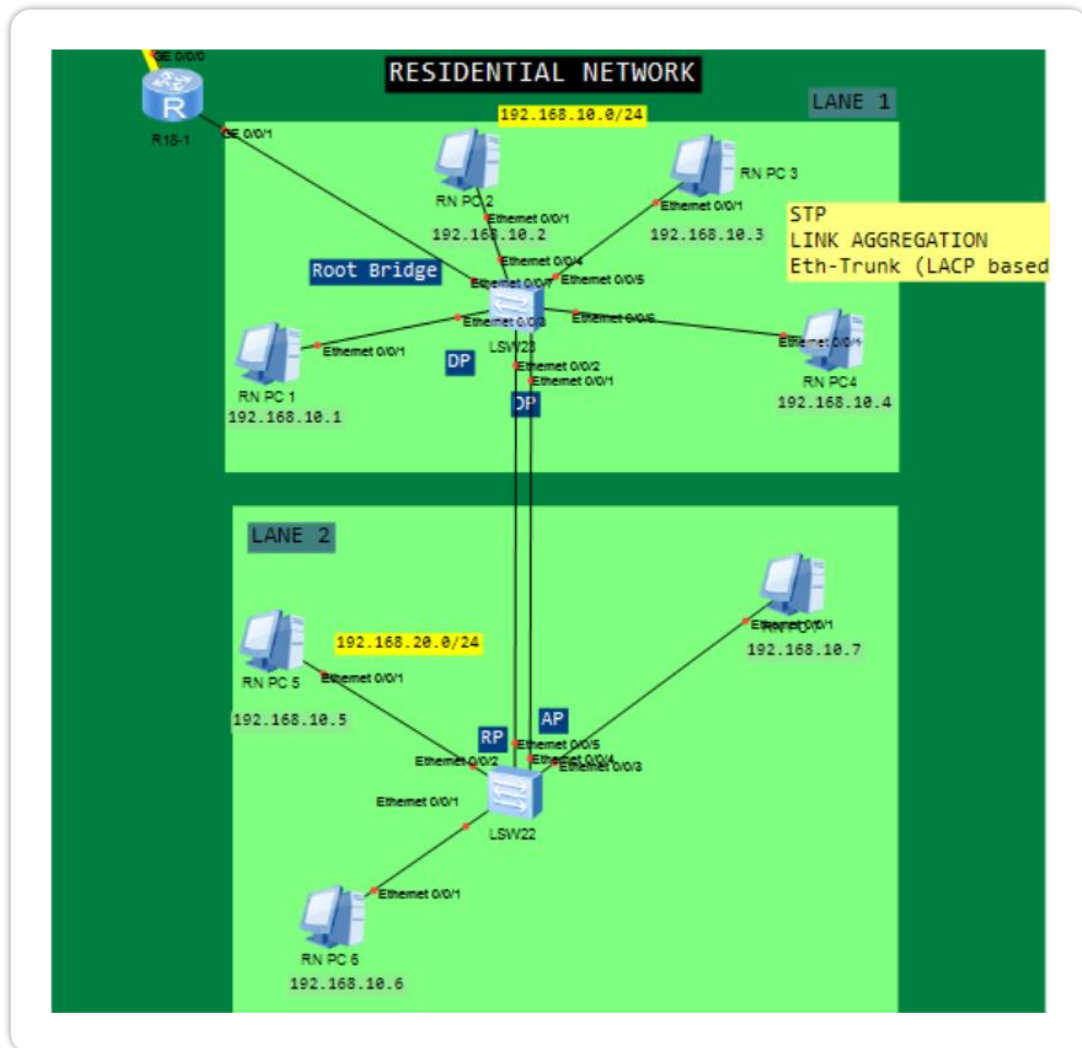
- aaa
 - o authentication-scheme default
 - o authorization-scheme default
 - o accounting-scheme default
 - o domain default
 - o local-user admin password cipher smart@123
 - o local-user admin service-type telnet

JUSTIFICATION

Implementing ACLs ensures that only authorized users can access sensitive areas of the network, thereby protecting against potential threats. VLANs contribute to better overall performance by reducing broadcast traffic, enhancing security through segmentation, and simplifying management tasks due to logically grouped entities despite physical locations. This setup ensures that resources are used efficiently while maintaining robust connectivity tailored specifically for the needs of your retail and commercial network.

RESIDENTIAL NETWORK

This diagram represents the structured layout of a retail and commercial network designed to support various business operations. The network is segmented into different areas, ensuring efficient resource management, secure communication, and streamlined operations across both residential and commercial environments.



Purpose

The purpose of this network is to ensure efficient data transmission, provide internet connectivity to multiple devices, enhance bandwidth through link aggregation, and maintain redundancy for uninterrupted service.

Components

1. Routers: 1
2. Switches: 2 LSW28 is the root bridge and its ports are Designated Ports.
3. PCs:7

Technologies Implemented

1. STP (Spanning Tree Protocol)
2. Link Aggregation
3. Eth-Trunk (LACP)

Description

1. **STP (Spanning Tree Protocol)**: Prevents network loops by creating a spanning tree within a mesh network of connected layer-2 bridges.
2. **Link Aggregation**: Combines multiple network connections in parallel to increase throughput beyond what a single connection could sustain.
3. **Eth-Trunk (LACP - Link Aggregation Control Protocol)**: Bundles several physical Ethernet links to form one logical link for load balancing and fault tolerance.

CONFIGURATION

SW1 Configuration

Eth-Trunk Configuration

- sys
- **Interface Eth-Trunk 1**
 - mode lacp
 - quit
- **Interface Ethernet 0/0/1**
 - eth-trunk 1
- **Interface Ethernet 0/0/2**
 - eth-trunk 1
 - quit

STP Configuration

- stp enable
- stp mode rstp
- quit

SW2 Configuration

Eth-Trunk Configuration

- sys
- **Interface Eth-Trunk 1**
 - mode lacp
 - quit
- **Interface Ethernet 0/0/4**
 - eth-trunk 1
 - quit
- **Interface Ethernet 0/0/5**
 - eth-trunk 1
 - quit

STP Configuration

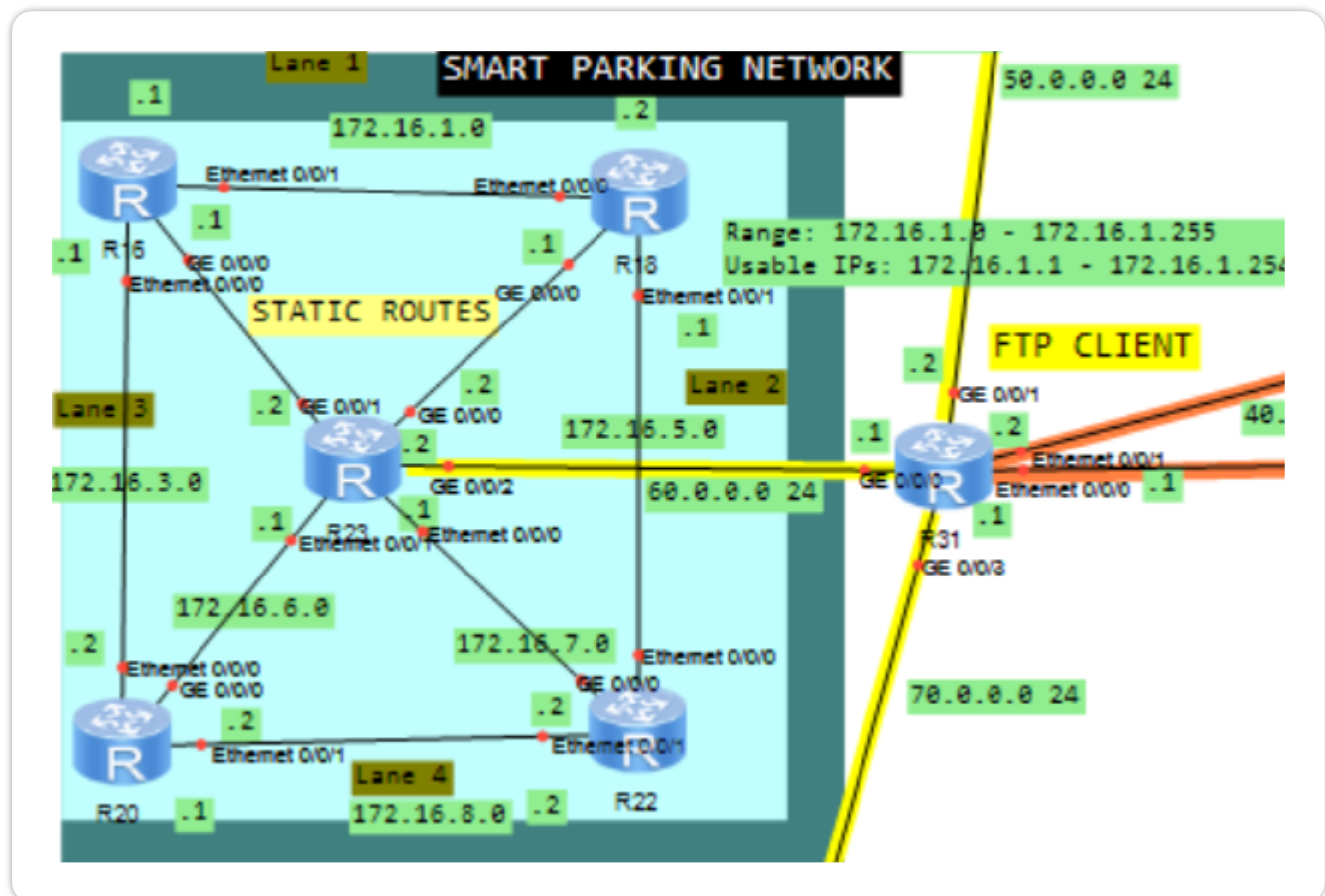
- stp enable
- stp mode rstp
- Info: This operation may take a few seconds. Please wait for a moment...done.
- stp root secondary
- quit

JUSTIFICATION

Implementing these technologies ensures optimal performance by preventing broadcast storms with STP, increasing available bandwidth through Link Aggregation, and providing link redundancy with Eth-Trunk using LACP for improved reliability. This setup creates a robust, secure, and efficient network infrastructure essential for supporting various services and applications within your retail and commercial network.

SMART PARKING NETWORK

This network represents the structured layout of a retail and commercial network designed to support various business operations. The network is segmented into different areas, ensuring efficient resource management, secure communication, and streamlined operations across both residential and commercial environments.



Purpose

The purpose of this network is to ensure efficient data transmission, provide internet connectivity to multiple devices, and optimize IP address allocation for scalability and performance.

Components

1. Routers: 5 every route shown between the router is represented as lane

Technologies Implemented

1. Static Route
2. VLSM (Variable Length Subnet Mask)

Description

1. **Static Route:** Manually added routes to a router's routing table to dictate where data should go within the network, providing precise control over traffic flow. In your network, static routes are used to manage traffic between different subnets, ensuring that data packets take the most efficient path to their destination.

Routes for each router:

ROUTER 16

```
ip route-static 172.16.5.0 255.255.255.0 172.16.1.2
ip route-static 172.16.4.0 255.255.255.0 172.16.1.2
ip route-static 172.16.7.0 255.255.255.0 172.16.1.2
ip route-static 172.16.8.0 255.255.255.0 172.16.1.2
ip route-static 172.16.6.0 255.255.255.0 172.16.1.2
```

```
ip route-static 172.16.4.0 255.255.255.0 172.16.2.2
ip route-static 172.16.7.0 255.255.255.0 172.16.2.2
ip route-static 172.16.8.0 255.255.255.0 172.16.2.2
ip route-static 172.16.6.0 255.255.255.0 172.16.2.2
ip route-static 172.16.5.0 255.255.255.0 172.16.2.2
```

```
ip route-static 172.16.8.0 255.255.255.0 172.16.3.2
ip route-static 172.16.6.0 255.255.255.0 172.16.3.2
ip route-static 172.16.7.0 255.255.255.0 172.16.3.2
ip route-static 172.16.2.0 255.255.255.0 172.16.3.2
ip route-static 172.16.5.0 255.255.255.0 172.16.3.2
ip route-static 172.16.4.0 255.255.255.0 172.16.3.2
```

ROUTER 18

```
ip route-static 172.16.2.0 255.255.255.0 172.16.1.1
ip route-static 172.16.3.0 255.255.255.0 172.16.1.1
ip route-static 172.16.6.0 255.255.255.0 172.16.1.1
ip route-static 172.16.7.0 255.255.255.0 172.16.1.1
```

```
ip route-static 172.16.8.0 255.255.255.0 172.16.1.1
```

ROUTER 22

```
ip route-static 172.16.1.0 255.255.255.0 172.16.3.1
```

```
ip route-static 172.16.5.0 255.255.255.0 172.16.8.2
```

```
ip route-static 172.16.4.0 255.255.255.0 172.16.6.1
```

ROUTER 22

```
ip route-static 172.16.1.0 255.255.255.0 172.16.5.1
```

```
ip route-static 172.16.3.0 255.255.255.0 172.16.8.1
```

```
ip route-static 172.16.2.0 255.255.255.0 172.16.7.1
```

```
ip route-static 172.16.4.0 255.255.255.0 172.16.7.1
```

```
ip route-static 172.16.5.0 255.255.255.0 172.16.6.1
```

ROUTER 23

```
ip route-static 172.16.1.0 255.255.255.0 172.16.4.1
```

```
ip route-static 172.16.3.0 255.255.255.0 172.16.2.1
```

```
ip route-static 172.16.8.0 255.255.255.0 172.16.6.2
```

```
ip route-static 172.16.7.0 255.255.255.0 172.16.7.2
```

2. **VLSM (Variable Length Subnet Mask):** Allows more efficient use of IP addresses by varying subnet mask lengths according to the number of hosts needed in each subnet. This technique is used in your network to optimize IP address allocation, reducing waste and allowing for scalable expansion. Each subnet is tailored to the specific needs of different segments, ensuring efficient use of available IP addresses.

CONFIGURATION

Router 16

- Enter System View
- `system-view`
- Enter system view, return user view with Ctrl+Z.
- Interface Configurations
 - `GigabitEthernet 0/0/2`
 - `interface GigabitEthernet 0/0/2`
 - `ip address 172.16.2.1 255.255.255.0`
 - `quit`
 - `GigabitEthernet 0/0/1`
 - `interface GigabitEthernet 0/0/1`
 - `ip address 172.16.3.1 255.255.255.0`
 - `quit`
 - `GigabitEthernet 0/0/0`
 - `interface GigabitEthernet 0/0/0`
 - `ip address 172.16.2.1 255.255.255.0`
 - `quit`

❖ **Static Routes**

- ip route-static 172.16.5.0 255.255.255.0 172.16.1.2
- ip route-static 172.16.4.0 255.255.255.0 172.16.1.2
- ip route-static 172.16.7.0 255.255.255.0 172.16.1.2
- ip route-static 172.16.8.0 255.255.255.0 172.16.1.2
- ip route-static 172.16.6.0 255.255.255.0 172.16.1.2
-
- ip route-static 172.16.4.0 255.255.255.0 172.16.2.2
- ip route-static 172.16.7.0 255.255.255.0 172.16.2.2
- ip route-static 172.16.8.0 255.255.255.0 172.16.2.2
- ip route-static 172.16.6.0 255.255.255.0 172.16.2.2
- ip route-static 172.16.5.0 255.255.255.0 172.16.2.2
-
- ip route-static 172.16.8.0 255.255.255.0 172.16.3.2
- ip route-static 172.16.6.0 255.255.255.0 172.16.3.2
- ip route-static 172.16.7.0 255.255.255.0 172.16.3.2
- ip route-static 172.16.2.0 255.255.255.0 172.16.3.2
- ip route-static 172.16.5.0 255.255.255.0 172.16.3.2
- ip route-static 172.16.4.0 255.255.255.0 172.16.3.2

• **Ping Test**

- ping 172.16.8.0

PING 172.16.8.0: 56 data bytes, press CTRL_C to break

Reply from 172.16.8.1: bytes=56 Sequence=1 ttl=255 time=70 ms

Reply from 172.16.8.1: bytes=56 Sequence=2 ttl=255 time=30 ms

Reply from 172.16.8.1: bytes=56 Sequence=3 ttl=255 time=30 ms

Reply from 172.16.8.1: bytes=56 Sequence=4 ttl=255 time=30 ms

Reply from 172.16.8.1: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 172.16.8.0 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 30/42/70 ms

Router 18

- Enter System View
- system-view
- Enter system view, return user view with Ctrl+Z.
- Interface Configurations
 - o GigabitEthernet 0/0/2
 - o interface GigabitEthernet 0/0/2
 - o ip address 172.16.1.2 255.255.255.0
 - o quit
 - o GigabitEthernet 0/0/1
 - o interface GigabitEthernet 0/0/1
 - o ip address 172.16.4.1 255.255.255.0
 - o quit
 - o GigabitEthernet 0/0/0
 - o interface GigabitEthernet 0/0/0
 - o ip address 172.16.5.1 255.255.255.0
 - o quit

❖ **Static Routes**

- ip route-static 172.16.2.0 255.255.255.0 172.16.1.1

- ip route-static 172.16.3.0 255.255.255.0 172.16.1.1
- ip route-static 172.16.6.0 255.255.255.0 172.16.1.1
- ip route-static 172.16.7.0 255.255.255.0 172.16.1.1
- ip route-static 172.16.8.0 255.255.255.0 172.16.1.1

Router 22

- Enter System View
- system-view
- Enter system view, return user view with Ctrl+Z.
- Interface Configurations
 - GigabitEthernet 0/0/1
 - interface GigabitEthernet 0/0/1
 - ip address 172.16.3.2 255.255.255.0
 - quit
 - GigabitEthernet 0/0/2
 - interface GigabitEthernet 0/0/2
 - ip address 172.16.6.2 255.255.255.0
 - quit
 - GigabitEthernet 0/0/0
 - interface GigabitEthernet 0/0/0
 - ip address 172.16.8.1 255.255.255.0
 - quit
- ❖ Static Routes
- ip route-static 172.16.1.0 255.255.255.0 172.16.3.1
- ip route-static 172.16.5.0 255.255.255.0 172.16.8.2
- ip route-static 172.16.4.0 255.255.255.0 172.16.6.1

Router 23

- Enter System View
- system-view
- sysname R20
- Interface Configurations
 - GigabitEthernet 0/0/0
 - interface GigabitEthernet 0/0/0
 - ip address 172.16.2.2 255.255.255.0
 - quit
 - GigabitEthernet 0/0/1
 - interface GigabitEthernet 0/0/1
 - ip address 172.16.4.2 255.255.255.0
 - quit
 - GigabitEthernet 0/0/2
 - interface GigabitEthernet 0/0/2
 - ip address 172.16.6.1 255.255.255.0
 - quit
 - GigabitEthernet 0/0/3
 - interface GigabitEthernet 0/0/3
 - ip address 172.16.7.1 255.255.255.0
 - quit
- ❖ Static Routes
- ip route-static 172.16.1.0 255.255.255.0 172.16.4.1
- ip route-static 172.16.3.0 255.255.255.0 172.16.2.1
- ip route-static 172.16.8.0 255.255.255.0 172.16.6.2
- ip route-static 172.16.7.0 255.255.255.0 172.16.7.2

JUSTIFICATION

Implementing static routes provides precise control over traffic flow, enhancing security and performance in a targeted manner. VLSM optimizes IP address usage, reducing waste and allowing for scalable expansion of the network as needed without major restructuring. This setup ensures that resources are used efficiently while maintaining robust connectivity tailored specifically for the needs of your retail and commercial network.

CONCLUSION

The network topology for the smart city infrastructure incorporates various technologies including Sub-netting, Telnet, Security (Access Control Lists - ACL), Eth-Trunk (Link Aggregation Control Protocol - LACP based), Static Routes, Routing Information Protocol (RIP), Spanning Tree Protocol (STP), Virtual LANs (VLANs), Router on a Stick (ROAS), Dynamic Host Configuration Protocol (DHCP), File Transfer Protocol (FTP), Open Shortest Path First (OSPF), AAA (Authentication, Authorization, and Accounting), Console Security, NDP/NDTP configurations. This setup supports efficient communication and data flow across healthcare, security and surveillance, utilities, smart parking, retail and commercial, smart building, and residential networks, ensuring robust and secure operations within the smart city environment.